

ANKIT GHOSH

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B.Tech, Artificial Intelligence
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Summary

do not mention mediapipeAI/ML Software Engineer specializing in **Deep Learning, Computer Vision**, and **Agentic AI** systems. Proven expertise architecting and deploying end-to-end production vision pipelines: curriculum-based LLM training (CIRCLE), custom object detection models (Veyra), automated image preprocessing, pose-based subject isolation, and quality optimization. Skilled in distributed inference orchestration (**Kubernetes, Docker**), containerized deployment, edge-device AI deployment for real-time inference, and production optimization for manufacturing-grade workloads.Strong technical foundation in **PyTorch, OpenCV**, data pipeline engineering, batch processing, and production infrastructure (**Docker, Kubernetes**). Experienced optimizing model lifecycle — from problem formulation and model design through training infrastructure to edge deployment at scale. Driven to bring AI/ML systems engineering expertise to SwitchOn's DeepInspect platform, enabling automated quality inspection across manufacturing operations globally.

Education

SRM IST-Kattankulathur B.Tech · Artificial Intelligence	2027 CGPA - 8.20/10
H.S.Memorial School 10th · Computer Science · Janai, West Bengal	Mar 2020 - Jun 2021 8.89
Swarajya Senior Secondary School 12th · Computer Science · Alwar, Rajasthan	Feb 2022 - Jul 2023 7.78

Experience

Sasken Technologies Link Intern · Software Developer · Full-Time	Dec 2025 - Present Sector V, Bidhannagar, Kolkata, within the Bengal Intelligent Park
- Developed a zero-shot conversational learning framework for a robotic system, enabling real-time knowledge validation and incremental learning from live speech input. - Built a multilingual speech-understanding and NLP pipeline supporting speaker-independent recognition, sentence normalization, and intent extraction, designed for production reliability across languages. - Integrated cloud-based LLM inference with on-device memory systems, balancing latency and context-awareness for real-time human-robot interaction.	
PRS Neurosciences https://prsneurosciences.com/ ML Intern · Intern · Full-Time	Jun 2025 - Present Virtual
- Built and shipped an automated image-preprocessing pipeline with pose-based subject isolation and background reduction, standardizing human-centric datasets across variable inputs. - Optimized storage and throughput using Guetzli/MozJPEG compression while preserving model-ready image quality. - Designed a modular, scalable single-image/batch processing workflow to improve downstream CV pipeline efficiency.	

Projects

CIRCLE | <https://github.com/ankitghoshthecreator/CIRCLE>

Jan 2026 - Present

Individual · PyTorch, Python, GPT-2 (base model), QLoRA (parameter-efficient fine-tuning), Knowledge Distillation (eval scoring), Groq (70B eval/critic agent, inference), DeepSeek 32B (local synthetic data generation), Docker, Kubernetes

- **Curriculum-based LLM training from scratch** — fine-tuned a GPT-2 base with PyTorch + QLoRA through a staged curriculum (raw English → summarization → grammar → punctuation-level semantics → multi-turn conversational understanding), using knowledge distillation for lightweight eval scoring at each stage.
- **Closed-loop synthetic data generation** — built an agentic eval pipeline where a 70B model (Groq-hosted) critiques the fine-tuned model's outputs after each training round, then autonomously writes targeted prompts instructing a local 32B DeepSeek model to generate new training data specifically addressing the weaknesses it found — turning evaluation directly into the next round's dataset.
- **Production-style orchestration** — containerized the full training/eval/data-generation loop with Docker and orchestrated it via Kubernetes, so each stage (generation, training, evaluation, feedback) ran as an independently scalable, restartable service rather than a monolithic script.

DynaFare | <https://github.com/ankitghoshthecreator/DynaFare>

Aug 2026 - Present

Individual · Java 17, Spring Boot, Python, FastAPI, XGBoost, PostgreSQL, Redis, Docker, Kubernetes, AWS, MCP

Spring AI

- Architected a distributed, AI-driven backend combining a Java/Spring Boot REST API with a Python/FastAPI ML inference service, applying SDLC best practices across requirements, design, implementation, and deployment for a production-shaped system.
- Exposed live pricing and ETA predictions through both REST APIs and MCP tools, enabling GenAI/agent-based consumption alongside standard client integration, backed by PostgreSQL for transactional data and Redis for low-latency feature lookups.
- Containerized services with Docker and deployed via Kubernetes manifests (Deployments, Services, HPA) on AWS-shaped infrastructure, with a Resilience4j circuit breaker ensuring graceful fallback and high availability under ML service latency or failure.

SageRAG — Enterprise Knowledge Assistant with Fine-Tuned LLM & Hybrid

May 2026 - Sep 2026

RAG | <https://github.com/ankitghoshthecreator/SageRAG>

Individual · Python, FastAPI, React, TypeScript, PostgreSQL, Redis, Qdrant, Docker, JWT, OCR

Machine learning, SDE

- Designed and developed a production-oriented Retrieval-Augmented Generation platform combining hybrid retrieval (dense vector + keyword search with cross-encoder reranking) and a parameter-efficient fine-tuned LLM (LoRA/QLoRA) to ground responses in private enterprise documents with source citations.
- Implemented an OCR-enabled document ingestion pipeline supporting PDF, DOCX, and scanned files, with intelligent chunking, metadata extraction, and duplicate detection to standardize heterogeneous inputs for downstream retrieval.
- Built a full-stack system with a FastAPI backend and React/TypeScript dashboard, implementing JWT-based authentication and role-based access control, with PostgreSQL, Redis, and Qdrant handling relational data, caching, and vector search respectively, containerized with Docker for reproducible deployment.

RepoGuard AI: Agentic Code Review Platform | <https://github.com/ankitghoshthecreator/RepoGuard-AI>

Dec 2025 - Feb 2026

Individual project · Python · LLM Agents · Docker · CI/CD

Agentic AI

- Built end-to-end CI/CD automation for build, test, and deployment processes using proven frameworks and technologies, auto-generating unit tests and PR summaries to improve software delivery quality and technical standards.
- Containerized the platform using Docker for reproducible, high-performance analysis across environments, applying analytical rigor to automate release testing workflows.
- Implemented a retrieval-based analysis layer combining static analysis and code structure validation, applying an eye for detail to reduce false positives and improve code review accuracy.

- Applied core software engineering principles to design and develop a high-volume, high-performance, high-availability payment processing system for the financial services industry, using a transaction state machine and double-entry ledger to deliver a lasting solution within budget and time estimates.
- Built a Redis-backed idempotency layer with strong analytical rigor and an eye for detail, maintaining data consistency under concurrent load, network retries, and service failures to meet technical standards for financial transaction integrity.
- Structured the system into API, transaction-engine, ledger, and repository layers using proven frameworks and technologies, applying clean REST API design for scalable, maintainable architecture.

Certifications

Programming in Java https://drive.google.com/file/d/1zhf_9a_gtrkgKoErrqI9Ibk773OAvJfw/view?usp=drivesdk NPTEL · NPTEL24CS105S252505015	Nov 2024
Data Base Management System https://drive.google.com/file/d/1skKhKAEoLOMXDhNYULJ65bPT-jH1eaUq/view?usp=drivesdk NPTEL · NPTEL25CS145S533211819	Sep 2025

Publications

Adaptive Portfolio Optimization using Robbins–Monro Stochastic Approximation: https://zenodo.org/records/17559546 zenodo	Nov 2025
• Developed an adaptive, time-series portfolio-optimization framework using Robbins–Monro stochastic approximation, enabling real-time weight updates under non-stationary market conditions. • Implemented volatility-aware forecasting via EWMA-based modeling, dynamically tuning risk penalty and learning-rate parameters to balance return maximization and risk control. • Evaluated the model on 10+ years of financial time-series data (Yahoo Finance), achieving a 22.8% annualized return and Sharpe ratio ≈ 1.0, with improved stability over static portfolio baselines.	

Competitions

Nitro Stack Hackathon https://nitrostack.ai/university/6a4e238c82556efe6ab3cb6f/certificate/6a6af3f8f77f4626aab0f425 1st Runner Up · NitroStack	Aug 2026
• Built and shipped a winning multi-agent MCP server automating industrial recovery — coordinating inventory risk assessment, supplier performance tracking, and automated purchase-order generation across a modular, decorator-driven architecture. • Engineered real-time machine health monitoring with anomaly detection and predictive maintenance scoring, triggering automated recovery workflows and technician dispatch from live equipment telemetry. • Designed a unified data-integration layer syncing ERP, MES, and IoT telemetry streams with a coordinator module orchestrating cross-module agent workflows, plus embedded Next.js UI widgets rendered natively inside MCP-compliant clients (Claude Desktop, NitroStudio).	

Skills

Programming Languages: Java, Python, GO, C/C++
Languages and Frameworks: Spring Data JPA, Spring Boot, Maven, Hibernate
Design and Simulation Tools: Docker, Unity C#, MCP
Technical Skills: Thymeleaf
Tools and Technologies: JUnit, OpenCV, Transformers, YOLO, pytorch, Mediapipe, Streamlit , React, TypeScript
Domain Knowledge: Natural Language Processing, Speaker Identification

Languages

English [Professional Working Proficiency], Bengali [Native Proficiency], Hindi [Native Proficiency], Russian [Limited Working Proficiency], Japanese [Elementary Proficiency]

Links

[GitHub](#), [LinkedIn](#), [Instagram](#), [LeetCode](#), [HackerRank](#), [Link](#)